[theorem]Corollary

Fixed Point Theorems

Chapter 5: Brouwer, Schauder, and Beyond

Based on Pathak's Introduction to Nonlinear Analysis (2018)

Outline I

Nonexpansive Mappings: Overview I

Definition

A mapping $T : K \to K$ on a nonempty subset K of a normed linear space X is said to be:

1. **Nonexpansive** if $\|Tx - Ty\| \leq \|x - y\|$ for all $x, y \in K$
2. **Contractive** if $\|Tx - Ty\| \leq \alpha\|x - y\|$ for some $0 < \alpha < 1$

Key Observation

Banach's contraction principle guarantees unique fixed points for contractive mappings, but nonexpansive mappings may not have fixed points without additional assumptions.

Example (Nonexpansive Without FPP)

On the unit circle $S^1 = \{x \in \mathbb{R}^2 : \|x\| = 1\}$, a rotation by angle $\theta \neq 2\pi k$ is nonexpansive but has no fixed point.

Pseudocontractive Mappings I

Definition (Definition 5.23)

Let K be a nonempty subset of a normed linear space X. A mapping $F : K \to X$ is said to be **pseudocontractive** if for each $k > 0$ and for all $x, y \in K$:

$$\|x - y\| \leq \|(1 + k)(x - y) - k(Fx - Fy)\|$$

Hilbert Space Characterization

In a real Hilbert space with corresponding norm, this is equivalent to:

$$(Fx - Fy, x - y) \leq \|x - y\|^2 \quad \text{for all } x, y \in K$$

Pseudocontractive Mappings II

Connection to Accretive Operators

Pseudocontractive mappings are closely related to accretive operators through the relationship:

$$T = I - F \text{ is accretive} \iff F \text{ is pseudocontractive}$$

Asymptotically Nonexpansive Mappings I

Definition (Definition 5.24)

Let K be a nonempty subset of a Banach space X. A mapping $T : K \to K$ is said to be **asymptotically nonexpansive** if:

$$\|T^n x - T^n y\| \leq k_n \|x - y\| \quad \text{for all } x, y \in K \quad \text{and} \quad \lim_{n \to \infty} k_n = 1$$

Key Properties

- More general than nonexpansive mappings
- Sequence of constants $\{k_n\}$ approaches 1
- Not all asymptotically nonexpansive mappings are nonexpansive

Historical Note

Goebel and Kirk proved fundamental theorems for this class in 1972, extending Banach's theory to this more general setting.

Asymptotic Regularity I

Definition (Definition 5.25)

A mapping $T : K \to K$ is said to be **asymptotically regular** if for any $x \in K$:

$$\|T^n x - T^{n+1} x\| \to 0 \quad \text{as } n \to \infty$$

Significance

Asymptotic regularity is a weaker condition than having a fixed point, yet it provides crucial control over the iterative behavior of the mapping.

Theorem (Bose, 1970)

Let X be a uniformly convex Banach space with weakly continuous duality mapping and K a nonempty, bounded, closed and convex subset of X. If $T : K \to K$ is asymptotically nonexpansive and asymptotically regular, then for any $x \in K$, the sequence $\{T^n x\}$ converges weakly to a fixed point of T.

Opial's Condition I

Definition (Definition 5.26)

A Banach space X is said to satisfy **Opial's condition** if for each $x_0 \in X$ and each sequence $\{x_n\}$ weakly converging to x_0, the inequality:

$$\liminf_{n\to\infty} \|x_n - x\| > \liminf_{n\to\infty} \|x_n - x_0\|$$

holds for all $x \neq x_0$.

Equivalent Form (Weak Opial's Condition)

If a sequence $\{x_n\}$ is weakly convergent to x_0 in X, then for every $x \in X$:

$$\limsup_{n\to\infty} \|x_n - x\| \geq \limsup_{n\to\infty} \|x_n - x_0\|$$

- Every Hilbert space and ℓ_p spaces $(1 < p < \infty)$ satisfy Opial's condition

Opial's Condition II

- Banach spaces with weakly continuous duality mappings satisfy weak Opial's condition
- Opial proved this inequality is strict for $x \neq x_0$

Convergence Theorem for Asymptotically Nonexpansive Maps I

Theorem (Theorem 5.43 - Bose, 1971)

Let X be a uniformly convex Banach space having weakly continuous duality mapping and K a nonempty, bounded, closed and convex subset of X. Suppose $T : K \to K$ is asymptotically nonexpansive and asymptotically regular. Then for any $x \in K$, the sequence $\{T^n x\}$ converges weakly to a fixed point of T.

Proof Outline

1. Every weak cluster point of $\{T^n x\}$ is a fixed point (by asymptotic regularity)
2. Opial's condition ensures all weak cluster points coincide
3. Uniform convexity provides the convergence framework

Convergence Theorem for Asymptotically Nonexpansive Maps II

Corollary (Corollary 5.12)

Let K be a closed, convex and bounded subset of X and let $T : K \to K$ be weakly continuous and asymptotically nonexpansive. If T is asymptotically regular at $x_0 \in K$, then $\{T^n x_0\}$ converges weakly to a fixed point of T.

Sequences of Asymptotically Nonexpansive Mappings I

Definition (Definition 5.27)

The sequence $\{T_n\}$ of self-mappings of K is said to be **asymptotically nonexpansive** if:

$$\|T_n x - T_n y\| \leq k_n \|x - y\| \quad \text{for all } x, y \in K \quad \text{with} \quad \lim_{n\to\infty} k_n = 1$$

Fixed Point Set Notation

We denote the set of fixed points of mapping T by $\mathcal{F}(T) = \{x \in K : Tx = x\}$.

Sequences of Asymptotically Nonexpansive Mappings II

Theorem (Theorem 5.44 - Pasty, extending Opial)

Let X be a uniformly convex Banach space with Fréchet differentiable norm and K a closed and convex subset of X. Let C be a subset of K and $S = \{T_n\}_{n=1}^{\infty}$ an asymptotically nonexpansive sequence of self-mappings of K such that:

1. $C \subset \bigcap_{n=1}^{\infty} \mathcal{F}(T_n)$
2. *There exists* $x_0 \in K$ *for which* $T_n x_0 \to z$ *implies* $z \in C$
3. $T_n T_m x_0 - T_n x_0 \to 0$ *as* $n \to \infty$ *for all fixed* m

Then either $C = \emptyset$ *and* $\|T_n x_0\| \to +\infty$, *or* $C \neq \emptyset$ *and* $T_n x_0$ *converges weakly to an element of* C.

The Fixed Point Property I

Definition (Definition 5.28)

A topological space X is said to possess the **fixed point property** (FPP) if every continuous function of X into itself has a fixed point.

Examples with FPP

- Closed unit interval $[0, 1]$ in $\mathbb{R}$
- Finite closed interval $[a, b]$
- Closed unit disk in the plane

Important Remarks

- It is difficult to establish FPP in general settings
- FPP is preserved under homeomorphisms (Theorem 5.45)
- Retracts play a crucial role in FPP theory

Partition of Unity I

Definition (Definition 5.29)

Suppose $V_1, \dots, V_n$ are open subsets of a locally compact Hausdorff space X, $K \subset X$ is compact, and:

$$K \subset V_1 \cup \cdots \cup V_n$$

Then for every $j = 1, \dots, n$ there exists $\varphi_j \in C(X)$, with $0 \leq \varphi_j \leq 1$, supported on V_j such that:

$$\varphi_1(x) + \cdots + \varphi_n(x) = 1, \quad \forall x \in K$$

The collection $\{\varphi_1, \dots, \varphi_n\}$ is called a **partition of the unity** for K subordinate to the open cover $\{V_1, \dots, V_n\}$.

Connection to Functional Analysis

The partition of unity is fundamental in constructing continuous functions and proving existence results in functional analysis.

Retracts and the Fixed Point Property I

Theorem (Theorem 5.46)

Any nonempty, closed and convex subset K of $\mathbb{R}^n$ or of a Hilbert space or of a Banach space X is a retract of any larger subset.

Theorem (Theorem 5.47)

If Y has the fixed point property and X is a retract of Y, then X has the fixed point property.

Proof Strategy

Let R be a retraction of Y onto X. If $F : X \to X$ is continuous, then $FR : Y \to X \subset Y$ is continuous. By FPP of Y, there exists $u \in Y$ such that $FRu = u$. If $u \in X$, then $Fu = u$. If $u \notin X$, we reach a contradiction.

Remark

These results are foundational for proving Brouwer's and Schauder's theorems using retraction arguments.

Brouwer's Fixed Point Theorem I

Theorem (Theorem 5.48 - Brouwer's FPT)

The closed unit ball $\mathbb{B}^n$ of $\mathbb{R}^n$ has the fixed point property. Equivalently: Let $f : \mathbb{B}^n \to \mathbb{B}^n$ be a continuous function. Then f has a fixed point $\bar{x} \in \mathbb{B}^n$.

Proof Technique: Differential Forms

Using the theory of differential forms and Stokes' theorem:

- Assume f has no fixed point
- Construct a continuous retraction $r : \mathbb{B}^n \to \mathbb{S}^{n-1}$ of class C^2
- Compute $\det(J_r(x)) = 0$ for all x, leading to a contradiction

Brouwer's Fixed Point Theorem II

Topological Proof

Alternative approach using topological degree and homology:

- Show that $\mathbb{S}^{n-1}$ is not a retract of $\mathbb{B}^n$ (topologically)
- Use homology to establish the contradiction

Extensions of Brouwer's Theorem I

Theorem (Theorem 5.49)

Every nonempty, compact and convex subset K of K of $\mathbb{R}^n$ (a finite dimensional normed linear space) has the fixed point property.

Proof.

For r sufficiently large, the ball B_r of radius r contains K. By Theorem 5.46, K is a retract of B_r. Since B_r is homeomorphic to $\mathbb{B}^n$, by Theorem 5.45, B_r has the FPP. Therefore, by Theorem 5.47, K has the FPP. $\square$

Theorem (Theorem 5.50)

Brouwer's FPT is equivalent to: there exists no retraction of $\mathbb{B}^n$ onto to the boundary $\partial\mathbb{B}^n$ which is continuously differentiable.

Extensions of Brouwer's Theorem II

Theorem (Theorem 5.51 - Consequence of Brouwer)

Let $F : \mathbb{B}^n \to \mathbb{R}^n$ be a continuous mapping and suppose that for some $r > 0$ and all $\eta > 0$:

$$F(x) + \eta x \neq 0 \quad \textit{for any } x \textit{ with } \|x\| = r$$

Then there exists a point x_0, $\|x_0\| \leq r$ such that $F(x_0) = 0$.

Schauder's Fixed Point Theorem I

Theorem (Theorem 5.55 - Schauder's FPT)

Let K be a nonempty, closed and convex subset of a normed linear space X. Let T be a continuous mapping of K into a compact subset of K. Then T has a fixed point in K.

Key Assumption

Unlike Brouwer, Schauder doesn't require finite dimensionality. Instead, the compactness of $T(K)$ provides the essential structure.

Schauder's Fixed Point Theorem II

Proof Strategy.

1. Let A be the compact subset with $T(K) \subset A \subset K$
2. A is contained in a closed, convex, bounded subset B of X
3. $T(B \cap K) \subset A \subset B$, which is compact
4. Use Brouwer's theorem on finite-dimensional approximations
5. Extract convergent subsequences from A's compactness

□

Schauder's Theorem: Extensions and Consequences I

Theorem (Theorem 5.56)

Let K be a nonempty, compact and convex subset of a normed linear space X and let T be a continuous mapping of K into itself. Then T has a fixed point in K.

Theorem (Theorem 5.57)

Let T be a compact and continuous mapping of a normed linear space X into itself and let $T(X)$ be bounded. Then T has a fixed point.

Theorem (Theorem 5.58)

Let X be a reflexive Banach space, K a closed and convex subset of X and T a weakly continuous mapping of K into a bounded subset of K. Then T has a fixed point K.

Schauder's Theorem: Extensions and Consequences II

Practical Implications

These extensions allow application to infinite-dimensional spaces when:

- The mapping is compact
- The space is reflexive with weak continuity
- The image has suitable boundedness properties

Relative Compactness Condition I

Theorem (Theorem 5.59)

Let X be a normed linear space. Suppose that K is a nonempty, closed, and convex subset of X, and T a continuous mapping of K into itself such that $T(K)$ is relatively compact in X. Then T has a fixed point.

Proof.

Let B be the convex hull of $T(K)$ in X, and let S be the closure of B in X. Then $S \subset K$ is:

- Nonempty and convex
- Compact (since $T(K)$ is relatively compact)
- Contains $T(K)$: $T(S) = T(\overline{B}) = T(\bar{B} \cap K) \subset \overline{T(K)} = S$

By Theorem 5.56, $T|_S$ has a fixed point. $\square$

Krasnoselski's Fixed Point Theorem I

Theorem (Theorem 5.61 - Krasnoselski)

Let K be a nonempty, complete and convex subset of a normed linear space X. Let T be a continuous mapping of K into a compact subset of X. Let $S : K \to X$ be a contradiction mapping with Lipschitz constant α and let $Tx + Sy \in K$ for all $x, y \in K$ for all $x, y \in K$. Then there is a point $u \in K$ such that $Tu + Su = u$.

Key Insight

This theorem combines:

- Compactness of one mapping (T)
- Contractiveness of another mapping (S)
- A closure property ($Tx + Sy \in K$)

Krasnoselski's Fixed Point Theorem II

Proof Idea.

Define $F(x, y) = Tx + Sy$ on $K \times K$. For fixed x, the map $y \mapsto F(x, y)$ is a contraction. Show that the mapping A defined through this process has a fixed point by Schauder's theorem. $\square$

Krasnoselski's Approximation Scheme I

Theorem (Theorem 5.62 - Krasnoselski Approximation)

Let K be a bounded, closed and convex subset of a uniformly convex Banach space X. Let T be a nonexpansive mapping of K into a compact subset of K. Let x_0 be an arbitrary point of K. Then the sequence defined by:

$$x_{n+1} = \frac{1}{2}(x_n + Tx_n), \quad n = 0, 1, 2, \ldots$$

converges to a fixed point of T in K.

Practical Algorithm

This provides a constructive method for finding fixed points:

1. Start with arbitrary $x_0 \in K$
2. Compute $x_{n+1} = \frac{1}{2}(x_n + Tx_n)$
3. Sequence converges to a fixed point

Krasnoselski's Approximation Scheme II

Convergence Proof.

- Show $\|x_{n+1} - y\| \leq \|x_n - y\|$ for any fixed point y
- Use uniform convexity and compactness
- Extract convergent subsequence

$\square$

Boundary Condition Theorems I

Theorem (Theorem 5.63 - Altman's Theorem)

Let X be a normed linear space. Let T be a continuous mapping of $B_r = \{x : \|x\| \leq r\}$ into a compact subset of X such that:

$$\|Tx - x\|^2 \geq \|Tx\|^2 - \|x\|^2, \quad \textit{for all } x \textit{ such that } \|x\| = r$$

Then T has a fixed point in B_r.

Theorem (Theorem 5.64 - Rothe's Theorem)

Let X be a normed linear space and T be a continuous mapping of B_r into a compact subset of X such that:

$$T(\partial B_r) \subset B_r, \quad \textit{i.e.}, \|Tx\| < \|x\| \quad \forall x \in \partial B_r$$

Then T has a fixed point.

Boundary Condition Theorems II

Key Difference

Rothe's condition is more restrictive (strict inequality) but easier to verify in practice than Altman's condition.

Potter's Fixed Point Theorem I

Theorem (Theorem 5.65 - Potter's Theorem)

Let X be a normed linear space. Let K be any closed and convex subset of X and T be a continuous mapping of K into a compact subset of X such that:

$$T(\partial K) \subset K$$

Then T has a fixed point.

Potter's Fixed Point Theorem II

Proof Sketch.

Define radial retraction R of X onto K by:

$$Rx = \frac{x}{\max(1, \rho(x))}$$

where $\rho(x) = \inf\{\alpha : x \in \alpha K\}$ is the Minkowski functional. Then:

- R is a continuous retraction
- RT maps K continuously into a compact subset of K
- By Schauder's theorem, RT has a fixed point u
- This fixed point is also a fixed point of T

$\square$

Finite-Dimensional Reduction I

Lemma (Lemma 5.8)

Let K be a nonempty, compact and convex subset of a finite dimensional real Banach space X. Then every continuous function $f : K \to K$ has a fixed point $\bar{x} \in K$.

Proof Outline.

Without loss of generality, assume X is homeomorphic to $\mathbb{R}^n$. We can assume $K \subset \mathbb{R}^n$.

1. Let $p(x) \in K$ be the unique point of minimum norm from x to K
2. Define $g(x) = f(p(x))$
3. g maps $\mathbb{R}^n$ continuously onto K
4. By Brouwer's theorem on finite dimensions, $g(\bar{x}) = \bar{x} = f(\bar{x})$

□

Finite-Dimensional Reduction II

Remark

[Remark 5.25] If there is a compact and convex set $K \subset \mathbb{R}^n$ such that $h(K) \subset K$, then h has a fixed point $\bar{x} \in K$.

Schauder-Tychonoff Fixed Point Theorem I

Theorem (Theorem 5.66 - Schauder-Tychonoff)

Let X be a locally convex space, $K \subset X$ nonempty and convex, $K_0 \subset K$ compact. Given a continuous map $f : K \to K_0$, there exists $\bar{x} \in K_0$ such that $f(\bar{x}) = \bar{x}$.

Schauder-Tychonoff Fixed Point Theorem II

Proof Strategy

1. Use partitions of unity to approximate f by finite-dimensional mappings
2. For each partition $\{V_1, \ldots, V_n\}$ of K_0, define:

$$f_U(x) = \sum_{j=1}^{n} \varphi_j(f(x))x_j$$

   where x_j are representative points
3. Apply Brouwer's theorem on finite-dimensional convex hull
4. Extract convergent subsequence using compactness of K_0

Remark

This is the most general fixed point theorem in the Brouwer-Schauder family, applicable to locally convex topological vector spaces.

Mann Iteration Algorithm

Mann Iteration Scheme

For a nonexpansive (or asymptotically nonexpansive) mapping T on convex set K:

$$x_{n+1} = (1 - \lambda_n)x_n + \lambda_n T(x_n), \quad n = 0, 1, 2, \ldots$$

where $\{\lambda_n\}$ is a sequence with $0 < \lambda_n < 1$ and $\sum \lambda_n(1 - \lambda_n) = \infty$.

```python
# Python: Mann Iteration Example
import numpy as np
import matplotlib.pyplot as plt

def mann_iteration(T, x0, lam, max_iter=100):
    """Mann iteration for fixed point computation"""
    x = np.copy(x0)
    trajectory = [x.copy()]

    for n in range(max_iter):
        Tx = T(x)
        x = (1 - lam) * x + lam * Tx
        trajectory.append(x.copy())

    return np.array(trajectory)

# Example: T(x) = 0.7*x + 0.5
```

Krasnoselski Iteration Example

Krasnoselski Method

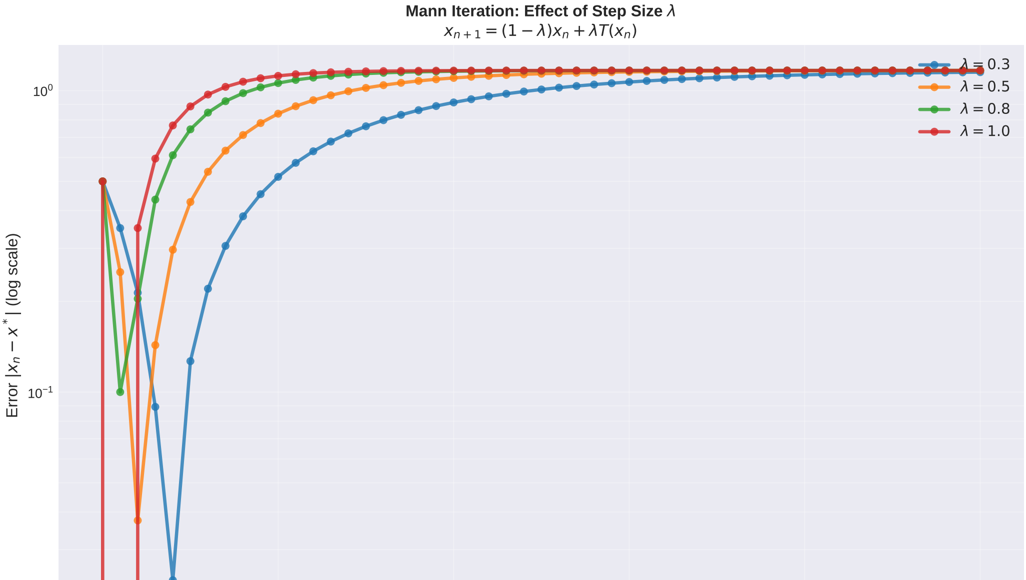
For nonexpansive mapping T on convex set K:

$$x_{n+1} = \frac{1}{2}(x_n + Tx_n), \quad n = 0, 1, 2, \ldots$$

This is Mann iteration with $\lambda_n = 1/2$.

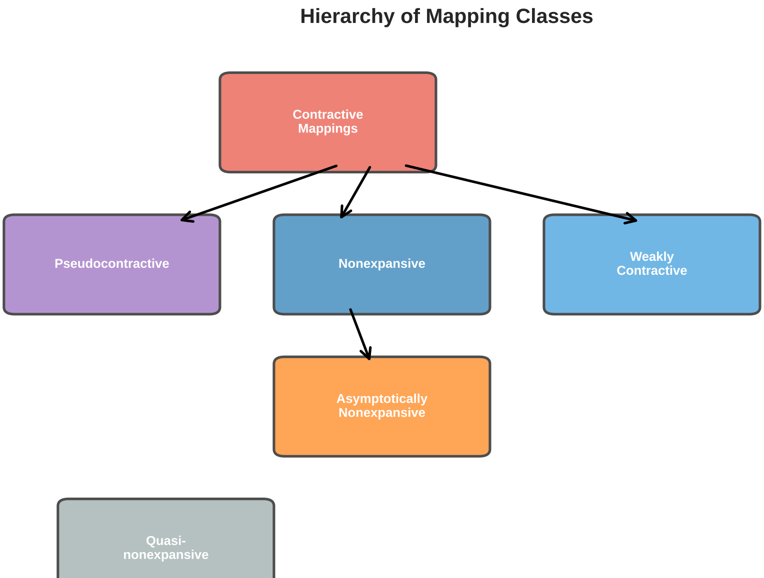
```
# Python: Krasnoselski Iteration Implementation
def krasnoselski_iteration(T, x0, max_iter=100, tol=1e-8):
    """Krasnoselski iteration (Mann with   =0.5)"""
    x = np.copy(x0)
    errors = []

    for n in range(max_iter):
        Tx = T(x)
        x_new = 0.5 * (x + Tx)
        error = np.linalg.norm(x_new - x)
        errors.append(error)

        if error < tol:
            print(f"Converged␣in␣{n+1}␣iterations")
            break
```

Convergence Behavior Visualization



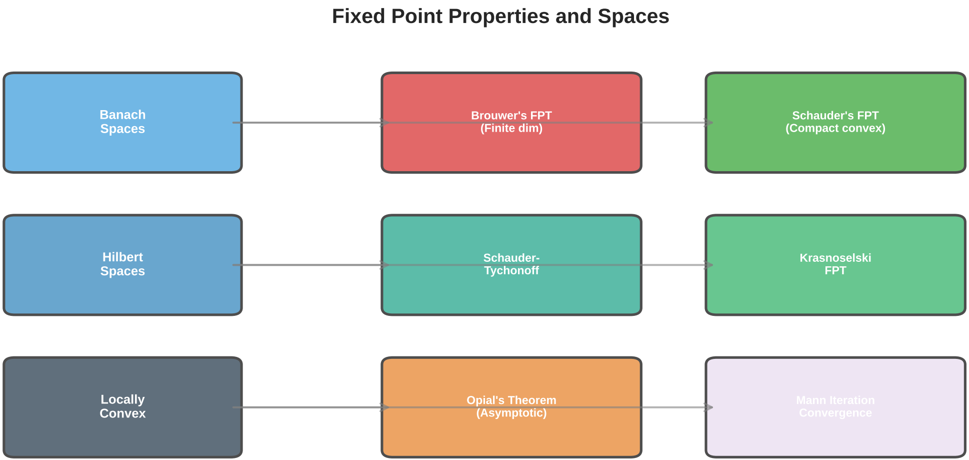
Mapping Hierarchy I

Mapping Hierarchy II



Fixed Point Spaces and Results I

Fixed Point Spaces and Results II



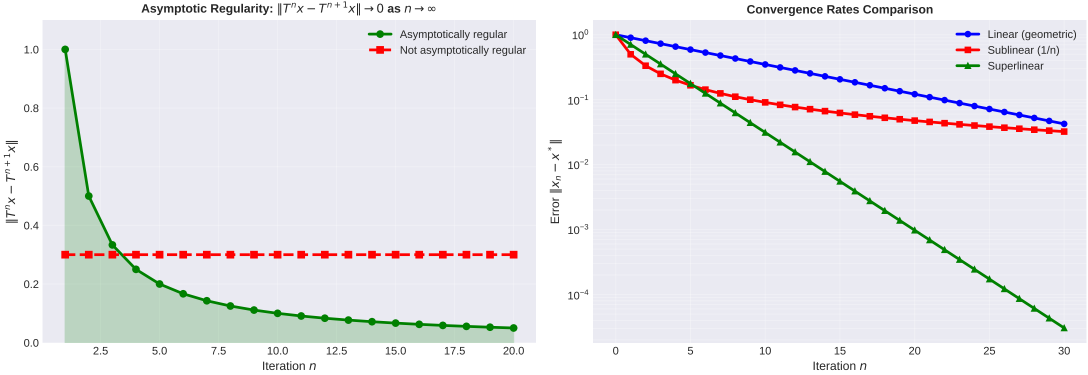
Theorem Conditions Summary I

Theorem Conditions Summary II

Fixed Point Theorems: Conditions and Conclusions

Theorem	Domain	Mapping Type	Key Condition	Conclusion
Banach	Metric Space	Contraction	$d(Tx, Ty) \leq \alpha d(x, y)$, $\alpha < 1$	Unique fixed point
Brouwer	$\mathbb{B}^n$	Continuous	Compact convex	Fixed point exists
Schauder	Banach	Compact map	$T(K) \subset$ compact $\subset K$	Fixed point exists
Schauder-	Locally convex	Compact map	Compact convex set	Fixed point exists
Tychonoff	topological			
Krasnoselski	Banach	$T + S$ sum	T compact, S contraction	Fixed point exists
Goebel-Kirk	Banach	Asympt. nonexp.	Bounded convex set	Fixed point exists

Asymptotic Regularity and Convergence I



Asymptotic Regularity and Convergence II

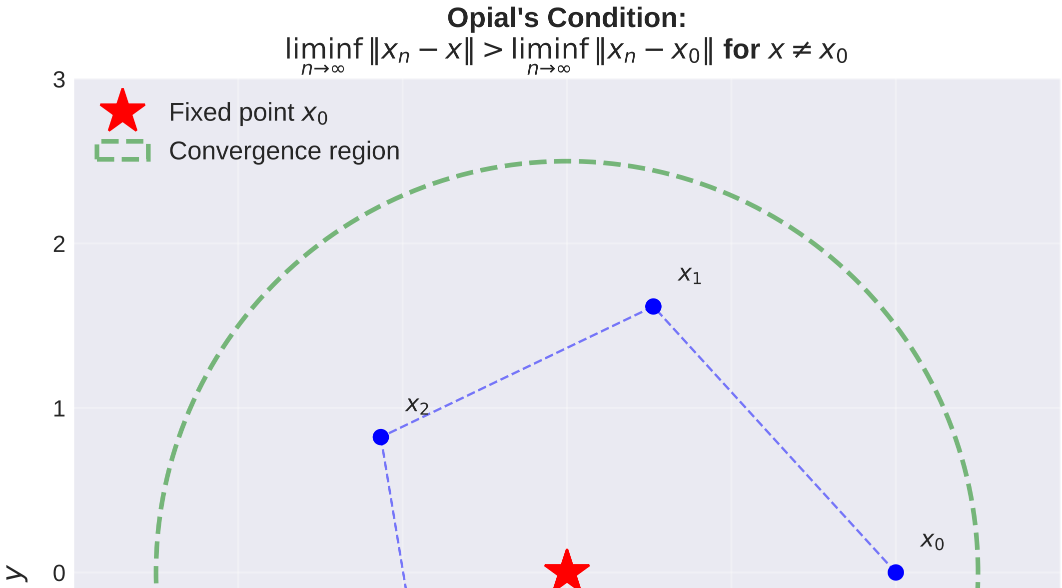
Interpretation

Left panel: Asymptotic regularity requires iterates to become closer, with $\|T^n x - T^{n+1}x\| \to 0$.
Right panel: Convergence rates vary dramatically:

- Linear (geometric): Fast, reliable
- Sublinear (1/n): Slow, guaranteed for asymptotic nonexpansive
- Superlinear: Very fast for special structures

Opial Condition Geometric Interpretation I

Opial Condition Geometric Interpretation II



Applications of Fixed Point Theorems I

Differential and Integral Equations

Fixed point theorems provide existence proofs for solutions to:

- Ordinary differential equations: $x'(t) = f(t, x(t))$
- Integral equations: $x(t) = g(t) + \int_a^t K(t,s)x(s)ds$
- Functional equations

Optimization

- Variational problems
- Saddle point problems
- Minimax theorems (related to fixed points)

Applications of Fixed Point Theorems II

Dynamical Systems

- Periodic orbits of dynamical systems
- Bifurcation analysis
- Invariant manifolds

Numerical Analysis

- Root finding via fixed point iteration
- Machine learning (fixed point methods in optimization)
- Control theory

Open Problems and Research Directions I

Known Open Questions

1. Does every nonexpansive mapping on a bounded closed convex set have a fixed point in all Banach spaces?
2. For which spaces does Opial's condition hold?
3. Optimal convergence rates for different mapping classes
4. Fixed point property for non-convex sets

Current Research Trends

- Multivalued mappings and set-valued analysis
- Fixed points in metric spaces and generalized metrics
- Applications to machine learning and optimization algorithms
- Variational inequalities and complementarity problems
- Fixed point theory for fractional operators

Remarks and Historical Context I

Historical Development

- **1886**: Poincaré's ideas on fixed points in dynamics
- **1904**: Bohl's retraction argument
- **1910**: Brouwer's elegant proof for $\mathbb{B}^n$
- **1930**: Schauder extends to infinite dimensions
- **1950s-70s**: Systematic study of nonexpansive and asymptotically nonexpansive mappings
- **Modern era**: Applications to functional analysis, PDE, and optimization

Key Insights

- Compactness and convexity are fundamental ingredients
- Topological degree and homology provide powerful tools
- Different spaces require adapted approaches
- Weak convergence methods extend classical results

Remarks and Historical Context II

Remark

Modern developments emphasize the interplay between topology, functional analysis, and computational methods in fixed point theory.

Key Takeaways

- Fixed point theorems guarantee existence under specific conditions
- Brouwer: finite-dimensional; Schauder: compact maps
- Asymptotic nonexpansivity allows weaker convergence
- Opial's condition and asymptotic regularity are technical tools
- Applications span PDE, optimization, and dynamics

"A fixed point is where the mapping rests; to find it is to understand the system."